A Trusted Silver Layer for Great Britain Road Collision Analysis

Bronze–Silver–Gold batch processing with PySpark

Group 16 | DAT535 | 11 November 2026

1. Problem, data and project scope

The project transforms Department for Transport (DfT) STATS19 data into reusable, validated analytical tables. Its main engineering contribution is the Silver layer: retaining the correct record grains, interpreting official codes and checking relationships before publication. A focused Gold analysis demonstrates how those tables can answer the following question:

> **Which lighting, road-type and speed-limit profiles account for the largest numbers of fatal or serious reported collisions in Great Britain, 2021–2025?**

The intended consumer is a road safety analyst choosing groups for closer local investigation. The analysis distinguishes absolute caseload, the proportion of collisions classified fatal or serious within a profile, and the profile's share of all severe collisions. These measures describe recorded collisions; traffic volumes and distance travelled are unavailable.

Source table	Records	CSV size (decimal MB)
Collisions	513,801	97.67
Vehicles	937,265	103.74
Casualties	652,821	53.69

The approximately 255 MB snapshot covers 2021–2025 and describes police-reported personal injury collisions in Great Britain. It excludes unreported events and does not represent every road accident. The official Excel data guide supplies category definitions. Original download dates were not recorded; validation timestamps must not be presented as acquisition dates. SHA-256 hashes identify the exact local input files, while mutable "last five years" URLs alone cannot reproduce a historical snapshot.

Architecture and reproducibility

Bronze preserves the three raw CSVs and reference guide. Silver publishes collision, vehicle and casualty Parquet tables plus a collision-level enriched table. Gold produces ten small summaries in Parquet and CSV, consumed by an executed notebook and generated figures. Scripts contain reusable transformations; notebooks serve interpretation and presentation.

The implementation uses Python 3.11, Spark 3.5.0 and Java 8 on the course VM. Separate development and production configuration files share the input snapshot but select different output directories. The repository contains source code, configuration, reference metadata and presentation evidence. Large raw inputs and generated layer directories are excluded from Git. Reproduction requires the externally retained CSVs whose hashes match the versioned baseline.

From the `project` directory, the main commands are:

```
bash scripts/run_tests.sh
bash scripts/run_project.sh -env dev
python scripts/render_results.py -env dev
```

The last command runs in the documented Python environment. Paths and Spark settings can be overridden without editing transformations.

2. Bronze and Silver: preserving meaning and grain

Bronze validation and provenance

Bronze acquisition is manual. The validator checks input presence, non-zero size, essential headers and duplicate column names, then records file sizes, SHA-256 hashes, column lists and change status. The reference guide is also hashed. Each attempt receives a distinct run directory and machine-readable report; the original CSVs remain unchanged. Checksums detect changed files, rather than duplicate records. Changed data still requires human review of schema, coverage and code definitions.

A relational Silver model

Table	Grain	Key
Collisions	One reported collision	`collision_index`
Vehicles	One vehicle within a collision	`collision_index`, `vehicle_reference`
Casualties	One casualty within a collision	`collision_index`, `casualty_reference`

A direct collision-level join of vehicles and casualties would multiply records: two vehicles and three casualties yield six joined rows. Instead, each child table is aggregated independently before joining its counts and attributes to collisions. The enriched table must retain exactly one row per collision. The original three tables remain available for other analytical questions, avoiding a flattened model tied to one report.

Targeted transformations

Numeric CSV types are inferred, with collision identifiers, reference strings and time explicitly preserved as text. This protects leading zeroes and avoids interpreting an `HH:mm` value as a timestamp. Dates are parsed into Spark dates; month, hour and weekend fields are derived. Conversion failures and inconsistencies against source year and weekday are counted.

Official labels are added for seven categorical fields: collision severity, casualty severity, weather, lighting, road surface, road type and vehicle type. The original codes remain available. Recognised unknown categories are distinguished from codes absent from the selected guide. Main severity codes remain distinct from enhanced severity categories; no severity adjustment is applied.

Age code -1 means “Data missing or out of range”. Keeping it in a numerical average would bias the result; deleting the whole record would lose useful information. Silver therefore preserves source ages and adds nullable clean columns. Negative ages become null there, while unexpected negative codes trigger review. Zero is valid for casualties. For example, $-1, 20, 40$ becomes null, 20, 40: the recorded-age average is 30, with a denominator of two, and all three records remain.

Similarly, a clean speed-limit column retains the six recognised public-highway limits, in mph. Missing or unknown source values remain traceable in the original field and become a separate missing group in Gold. Missing coordinates are counted rather than used as a blanket reason for rejection.

Why these choices matter

A null-only audit would miss populated special codes such as -1. Interpretation of the reference guide is therefore essential. Cleaning is deliberately limited to justified transformations: values already suitable for analysis are retained. This makes information loss explicit at the attribute level, while preserving reusable records and their source vocabulary.

3. Quality evidence, tests and publication

The reference development run is `20261110T150552746288Z`. Its Silver report passed all implemented gates. Input and output counts agree at every source grain:

Table	Input	Output	Missing keys	Duplicate excess
Collisions	513,801	513,801	0	0
Vehicles	937,265	937,265	0	0
Casualties	652,821	652,821	0	0

There were no unmatched collision relationships, casualty-to-vehicle orphans, parent/child year mismatches, or discrepancies between observed child counts and source collision counters. There were no date/time conversion failures, calendar inconsistencies, unsupported speed values or unmapped codes in the seven decoded fields. All four Silver Parquet tables were reloaded and checked for row-count and data-type agreement.

The report also records limitations rather than hiding them: 142,240 driver ages and 14,106 casualty ages use the missing-age code; 53 longitudes and 53 latitudes are absent; three speed limits are coded missing. Recorded ages span 1–105 for drivers and 0–105 for casualties. These extrema are observations, not independent validation of each age. Coordinate range checks establish global plausibility, not geographic accuracy. No source records were removed.

Complementary checks and failure behaviour

Key checks enforce table grain. Foreign-key checks detect orphans. Comparing observed children with collision counters tests completeness relative to the parent's declarations. Checking enriched collision counts guards against join multiplication. These checks address different failure modes; none alone establishes complete data correctness.

Missing or duplicate keys and empty source tables cause failure. Relationship anomalies, unsupported categories and other implemented quality violations block Silver publication. A run requiring review does not proceed to Gold. Reports are written before successful pointers are replaced atomically, and failed outputs remain inspectable. A file lock prevents simultaneous full runs using the same operational directory.

Publication is atomic per pointer, not a transaction spanning every layer. Silver can advance before a later Gold failure. Consumers needing a consistent end-to-end snapshot should use the pipeline manifest and its run identifiers. Standalone stages should not run concurrently against shared outputs.

Tests and observed execution

The saved test evidence records **17 successful tests**. Tiny fixtures exercise identifier preservation, keys, guide decoding, date parsing, missing ages, enrichment grain, overlapping vehicle groups, source changes, analytical denominators and publication. In the failed-update integration test, a vehicle is removed: Silver requires review, Gold is skipped and the previous validated Silver/Gold pointers remain unchanged. Tests require the small versioned guide but no large raw CSVs.

The reference full local run took 81.37 seconds: Bronze 0.95 s, Silver 59.49 s and Gold 19.48 s, plus orchestration overhead. A retained production-directory manifest, `20261110T180208676518Z`, also reports a passed full run (74.00 s). This is local operational evidence; the manifest alone does not establish which GitHub event launched it or confirm repository protection settings.

4. Gold results: answering the analytical question

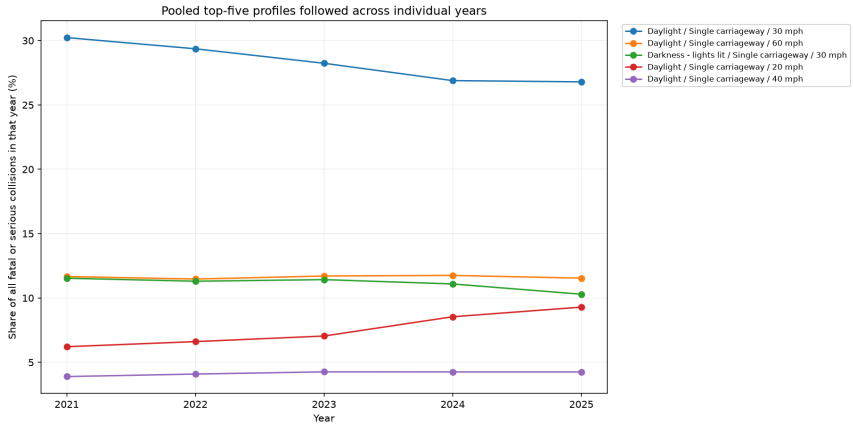
Gold contains 180 pooled lighting–road-type–speed-limit profiles and 803 observed annual profile combinations. Together with monthly, hourly, weather, lighting, road-type, speed-limit, weekend and vehicle-type tables, these make ten outputs. Exclusive summaries reconcile to Silver globally; annual profiles also reconcile within each year. Vehicle-type memberships overlap: a collision involving a car and cycle belongs to both groups, so those counts must not be summed.

Daylight / single carriageway / 30 mph accounts for the largest recorded severe caseload: 35,130 fatal or serious collisions, or 28.247% of all severe collisions. The five leading profiles contain 78,008 severe collisions, representing 62.725% of the total. Across the snapshot, 124,366 of 513,801 collisions are fatal or serious (24.205%).

Profile (all on single carriageways)	Severe cases	Within profile	Share of severe
Daylight / 30 mph	35,130	22.512%	28.247%
Daylight / 60 mph	14,458	35.502%	11.625%
Darkness, lights lit / 30 mph	13,817	27.544%	11.110%
Daylight / 20 mph	9,436	20.699%	7.587%
Daylight / 40 mph	5,167	27.957%	4.155%

The leading profile's within-profile severe proportion, 22.512%, is below the overall 24.205%. Its share of all collisions is 30.371%, compared with 28.247% of severe cases: a gap of −2.124 percentage points. Its large severe caseload accompanies a large recorded collision population.

By contrast, darkness without lighting / single carriageway / 60 mph contains 5,006 severe cases, with 37.125% classified severe within the profile. Its shares of all collisions and severe cases are 2.624% and 4.025%, respectively. It has a higher severe proportion but a smaller absolute caseload. This illustrates why counts and proportions answer different questions.



The same top five profiles retain ranks 1–5 in the same order in every year from 2021 to 2025. The leading profile has a negative severe-minus-all share gap in all five years; the unlit 60 mph comparison has a positive gap in all five. Each year uses its own denominator. This supports descriptive consistency within the observed period, rather than independent validation. Profiles were selected using the pooled data, and reporting changes remain relevant.

An analyst can use these groups to select local investigations, then examine locations, traffic exposure and infrastructure context. These results cannot rank risk per kilometre or determine whether changing lighting or speed limits would prevent severe collisions.

5. Performance, delivery and critical reflection

Spark implementation and measured performance

Transformations use built-in DataFrame operations; the monthly summary uses Spark SQL. Python row UDFs are unnecessary. Source frames are cached because audits, joins and writes reuse them, then unpersisted. Eight shuffle partitions are a small-VM starting point rather than a demonstrated optimum. Saved plans expose aggregations, scans, joins and shuffles. Silver is not partitioned into many year/month directories because this snapshot is modest; small Gold tables use one Parquet file each.

A separate experiment compares the same three-column yearly aggregation on CSV and Parquet. It uses an explicit CSV schema, no Spark cache and three trials in alternating format order. Query results match in every trial.

Format	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Median (s)
CSV	6.0336	1.0380	0.7430	1.0380
Parquet	1.4044	0.3758	0.2923	0.3758

The Parquet plan shows a batched, three-column scan. Both formats use partial aggregation and a shuffle. These measurements support Parquet for this workload, but uncontrolled JVM warm-up, OS page cache and VM load prevent a general speedup or scalability claim. The benchmark references an earlier validated run on the same input snapshot. An exit-137 failure during concurrent Spark activity also motivated sequential runs and conservative driver-memory settings on the 8 GB VM.

CI/CD and operational delivery

The project is merged into `main` at commit `e79206d`. The development workflow runs fixture tests on pushes to `dev`, same-repository pull requests targeting `dev`/`main`, and manual dispatch. Fork pull requests are excluded from the self-hosted test job. Test logs are retained as artifacts; legacy labs remain optional manual jobs.

The current production workflow runs on pushes to `main` or manual dispatch, executes tests followed by Bronze–Silver–Gold against the VM's external snapshot, then generates presentation artifacts. It uses a persistent production directory and the `main` environment. **The current YAML has no monthly schedule.** Environment approvals and required branch checks depend on GitHub settings and cannot be inferred from YAML or the merge. A retained passed production manifest demonstrates a run in the production directory; remote run URLs and approval evidence would separately establish the hosting lifecycle.

Reflection and conclusion

The strongest engineering choice is preserving relational grain while making publication depend on measured quality. Official-code interpretation prevents misleading age calculations; separate child aggregation prevents inflated collision counts. Keeping raw codes and nullable prepared fields improves traceability, although downstream users must still choose valid denominators.

This is a full-snapshot batch pipeline on one VM, not an incremental or multi-node platform. Priorities for expansion are explicit schema contracts, archived input versions, source-change review, retention policies and resource sizing based on measured workloads. The guide reader is tied to the selected workbook layout. Unknown acquisition dates, reporting bias, unadjusted severity and missing traffic exposure remain material limitations.

Silver provides reusable, structurally validated data; Gold demonstrates a defensible application. The leading severe-case groups are consistent across the five observed years, while case volume and within-profile severity remain distinct. The result supports targeted investigation with clear boundaries on interpretation.

Sources and evidence. DfT, Road safety open data; DfT, Severity adjustment guidance. Repository evidence: analytical results and performance manifests in the reports directory, the executed analysis notebook and per-run quality reports. Contains public sector information licensed under the Open Government Licence v3.0.